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DeepCausality: A Hypergeometric Computational Causality Library for Rust

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Summary

DeepCausality is a hypergeometric computational causality library written in Rust, designed to build systems that reason about cause and effect. Hosted by the Linux Foundation for Data & AI since 2023, it supports uniform reasoning across deterministic and probabilistic modalities, including static and dynamic contextual causal models. At its core, the library implements the paradigm of "Causality as a spacetime-agnostic functional dependency", inspired by the causaloid formalism of Hardy (2005), using **Causaloids** (self-contained causal units) and **Contexts** (hypergraph environments). It connects abstract causal reasoning, physical laws, and programmable ethics in a single toolset for researchers and engineers building complex, verifiable autonomous systems.

Statement of need

Causal inference toolkits are usually specialized along one axis at a time: a library handles Markovian dynamics *or* non-Markovian memory, deterministic structural models *or* probabilistic graphical models, static causal graphs *or* time-varying ones, offline discovery from datasets *or* online reasoning inside a running system. Combining these regimes in a single model—say, a deterministic physical law nested inside a probabilistic Markov layer that itself sits inside a non-Markovian, context-dependent envelope, evaluated at runtime under uncertainty—typically requires gluing several disparate packages together, with no shared type system and no shared semantics for composition.

DeepCausality is, to the authors' knowledge, among the first packages to treat these regimes as orthogonal axes of a single framework rather than as separate libraries. Within one type-safe Rust API it supports: (i) **deterministic and probabilistic** modalities (the latter via `Uncertain<T>` first-class uncertainty propagation), (ii) **static and dynamic** causal structure (Causaloids over fixed graphs or over time-evolving contexts), (iii) **Markovian and non-Markovian** processes (the Causal Monad threads explicit state, context, and history through `bind`, so memory effects are expressible without leaving the abstraction), and (iv) **offline discovery and online reasoning** (a Causal Discovery Language for fitting models from data and a `PropagatingEffect` runtime for executing them inside deployed software). Because the Causaloid is recursive and isomorphic, these regimes can be mixed at arbitrary granularity inside a single model.

Real systems rarely sit in only one regime. A flight-envelope monitor mixes deterministic aerodynamics, probabilistic sensor models, non-Markovian fatigue history, and dynamic context. A tumor-treatment model mixes deterministic pharmacokinetics, probabilistic response, and non-Markovian patient history. DeepCausality provides a single, high-performance, type-safe framework in which such models can be expressed, composed, intervened on, and executed in real time—rather than forcing the practitioner to re-implement bridging code between several incompatible toolkits.

Software design

DeepCausality is structured as a monorepo containing 20 crates, organized into five layers (Figure 1).

```
┌──────────────────────────────────────────────────────────┐
│  Causal Discovery     deep_causality_discovery           │
├──────────────────────────────────────────────────────────┤
│  Causal Framework     deep_causality_core                │
│                       deep_causality                     │
│                       deep_causality_ethos               │
├──────────────────────────────────────────────────────────┤
│  Physics              deep_causality_physics             │
├──────────────────────────────────────────────────────────┤
│  Mathematics          deep_causality_tensor              │
│                       deep_causality_multivector         │
│                       deep_causality_topology            │
├──────────────────────────────────────────────────────────┤
│  Foundation           deep_causality_num                 │
│                       deep_causality_haft                │
│                       deep_causality_metric  ── shared sign conventions
└──────────────────────────────────────────────────────────┘
```

Figure 1: Layered architecture of DeepCausality. Upper layers depend on lower layers; the foundation layer (`deep_causality_metric`) supplies shared sign conventions used uniformly across mathematics, physics, and the causal framework.

Foundational layer. *HAFT* implements Higher-Kinded Types (HKT) in Rust via a witness pattern, providing Functor, Applicative, and Monad traits across container types. *NUM* defines an algebraic hierarchy (from Magma to Division Algebras) and specialized numeric types including `DoubleFloat` (double-double precision), `Complex`, `Quaternion`, and `Octonion`. *Rand* provides random number generation and statistical distributions for stochastic simulations.

Dynamic causality. The main `deep_causality` crate models causality via Causaloids that compose into causal graphs. The `deep_causality_core` crate defines the **Causal Monad** (`CausalMonad`), realized as `PropagatingEffect` (stateless) and `PropagatingProcess` (stateful) types that thread state, context, error, and audit logs through monadic `bind`. It also provides a `ControlFlowBuilder` for constructing correct-by-construction static execution graphs, with optional zero-allocation, ZST-only enforcement via the `strict-zst` feature, suitable for safety-critical, hard real-time, and `no_std` environments. *Discovery* provides a type-safe builder pipeline for the Causal Discovery Language (CDL).

Physics & metrics. *Physics* is a standard library of physics kernels and causal wrappers organized by domain— Astrophysics, Quantum Mechanics, Electromagnetism, Relativity, Thermodynamics—leveraging Geometric Algebra and Gauge Fields (Furey, 2024; Tong, 2018) and drawing on recent results in quantum geometry (Haruna, 2025; Kang et al., 2024). *Metric* defines foundational signatures for consistent handling of geometric properties.

Data structures. *Topology* implements `Graph` (sparse-matrix based), `Hypergraph`, `SimplicialComplex`, and `Manifold`, enabling TDA algorithms such as Vietoris–Rips triangulation. *Tensor* and *Sparse* provide N-dimensional `CausalTensor` support with Einstein summation and CSR matrices. *MultiVector* implements Clifford Algebra for relativistic geometry. *Ultragraph* provides a high-performance hypergraph backend (Liu et al., 2022).

Research & applications. *Algorithms* implements **SURD** (Synergistic, Unique, and Redundant decomposition) (Martínez-Sánchez & Lozano-Durán, 2025) and **mRMR** feature selection (Zhao et al., 2019). *Ethos* introduces a programmable deontic logic layer (`Teloid`) (Olson et al., 2024) for verifying proposed actions against operational objectives. *Uncertain* provides

Uncertain<T> (Bornholt et al., 2014), representing values as probability distributions and propagating uncertainty through computations.

Causaloid and Causal Monad: structure and sequencing on equal footing

Most existing causal frameworks force a choice between two incompatible shapes. Either the model is a **structural artifact**—a DAG, a structural causal model, a Bayesian network—which captures rich causal topology but offers no native story for functional sequencing or side-effect discipline; or the model is a **sequenced pipeline**—a chain, a stream, a monadic computation—which composes cleanly but flattens causal structure into a linear or tree-shaped flow. Practitioners typically pick one and pay for the other in glue code.

DeepCausality refuses that compromise. The **Causaloid** is the isomorphic, recursive structural primitive: it can be an atomic causal unit, a causal collection, or a causal graph, and because a Causaloid-of-Causaloids is itself a Causaloid, arbitrarily complex causal relations—nested graphs, hierarchies, hypergraphs of sub-models—can be expressed without leaving the type. The **Causal Monad** is the sequencing primitive: it enforces strict functional composition, ordered evaluation, state and context propagation, error handling, and audit logging via monadic `bind`. Neither is built on top of the other; they are co-equal abstractions targeting orthogonal concerns.

The two meet through `PropagatingEffect` (Figure 2).

```
        Causaloid                              Causal Monad
   (structural primitive)                 (sequencing primitive)
   ─────────────────────                  ─────────────────────
   • atomic | collection | graph          • PropagatingEffect
   • recursive: Causaloid-of-Causaloids   • PropagatingProcess
   • isomorphic nesting                   • bind / state / context
   • arbitrary causal topology            • error / audit logging
            │                                       │
            │     evaluates to / consumed by        │
            └──────────────┬────────────────────────┘
                           ▼
                   PropagatingEffect
                  (shared value type)
                           │
                           ▼
        freely interleavable: DAG inside pipeline,
        pipeline inside DAG, at any granularity
```

Figure 2: Causaloid and Causal Monad are co-equal primitives targeting orthogonal concerns (structure vs. sequencing). They meet through `PropagatingEffect`, the shared value type produced by Causaloid evaluation and consumed by monadic `bind`, which makes structural and sequenced models freely interleavable. A Causaloid's evaluation produces a `PropagatingEffect`, which is also the value carried by the Causal Monad's `bind`. As a result, structure and sequencing become freely composable: an arbitrarily complex causal graph can be lifted as a single monadic step, a monadic pipeline can sit inside a Causaloid, and the two can be interleaved at any granularity. A user can build a pure DAG of Causaloids, a pure monadic pipeline, or—most usefully—a mixed model in which functionally sequenced stages each contain rich causal sub-structure. This is the property that makes the multi-physics pipelines in `examples/physics_examples` (e.g., `gauge_gr`, `quantum_geometric_tensor`, `gravitational_wave`, `multi_physics_pipeline`) compose end-to-end: each stage is a sequenced Causal Monad step, and each step internally evaluates a Causaloid that may itself contain a graph of further Causaloids.

Algebraic and type-level integration

Two additional mechanisms keep the ecosystem coherent. First, core data structures (`CausalTensor`, `CausalMultiVector`, `Manifold`) are generic over numeric types implementing `Field` or `RealField` traits from `deep_causality_num`, which decouples algorithms from numeric representation and lets the 106-bit `DoubleFloat` type drop into all tensor and topological operations without code changes. Second, the witness-pattern HKT in `deep_causality_haft` makes heterogeneous types (a `Manifold`, a `CausalTensor`, a `CausalMultiVector`) implement the same `Functor`/`Monad` traits, so they interoperate within one monadic flow—the basis on which the physics crate is built.

Research impact statement

DeepCausality has been hosted by the Linux Foundation for Data & AI since 2023, with public development history spanning roughly three years from the initial commit on 2023-06-18 to the present, accumulated through sustained, iterative work rather than a single bulk drop. The library supports work in: **causal AI safety** (the `Ethos` layer enables formal verification of agent actions against safety protocols); **algorithm development** (open-source SURD and mRMR implementations for decomposing causal signals in high-dimensional data); **geometric deep learning** (Topology and MultiVector algebra as geometric priors); and **physics-informed causal modeling** (digital twins that are both causally sound and physically valid).

The repository ships example crates covering classical causality, contextual structural models, chronometric reasoning, avionics (`flight_envelope_monitor`, `geometric_tcas`, `magnav`), medicine (`aneurysm_risk`, `epilepsy`, `protein_folding`, `tumor_treatment`), material science, and physics (`gauge_gr`, `quantum_geometric_tensor`, `gravitational_wave`, `multi_physics_pipeline`).

Adoption and downstream use. DeepCausality is used in production by the ServiceRadar observability platform (https://github.com/carverauto/serviceradar), which describes its causal engine as providing "real-time triage and isolation via DeepCausality" over heterogeneous telemetry. It is also the core dependency of the research program at the Center for Dynamic Causality, which has produced multiple preprints that build on DeepCausality primitives (Causaloid, Causal Monad, `PropagatingEffect`, the physics and topology crates). A running list of project preprints, publications, talks, and technical write-ups is maintained at https://www.causalcenter.com/publications/.

Open development practices

The project is developed openly on GitHub under the MIT license. All 20 crates are published on crates.io with documentation on docs.rs; releases are tagged and archived on Zenodo for DOI-based citation. The public issue tracker and pull request workflow are open to external contributors, and a Developer Certificate of Origin (DCO) governs code contributions. Continuous integration runs the full test matrix on every push (build, test, lint via `make build`, `make test`, `make fix`), and each crate carries its own README, CHANGELOG, and SBOM artifact. Governance is documented in `MAINTAINERS.md`, `CODEOWNERS`, `CONTRIBUTING.md`, `CODE_OF_CONDUCT.md`, and `SECURITY.md` at the repository root.

AI usage disclosure

The DeepCausality project uses generative AI for software specification, code review, and testing and QA assistance. Tools used include Google Gemini Pro 3.1, Anthropic Claude Opus 4.6, and Claude Opus 4.7. All code design and architectural decisions were made by the human author, and all AI-assisted outputs were reviewed, edited, and validated by the human author before being incorporated into the codebase, documentation, or this manuscript.

Conflict of interest

The author declares no conflict of interest.

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